

# MGT 403 Business Analytics for Managers

Fall 2025

**Prof. Robert Montgomery**

## DESCRIPTION

This course provides a foundation for business analytics, and is intended for students of all backgrounds. Students will gain experience processing data, constructing mathematical models to improve decision making, and using statistics to gain business insights. Data analytical fluency has become an essential skill for any manager. As such, we will work extremely hard to ensure students have all the support and resources they need to succeed in this course. This course will also serve as a bridge to higher level analytics courses within The Rady School of Management.

The course will be comprised of three units: Statistic & Uncertainty, Decision Analysis, and Optimization. The first unit will focus on data visualization and probability, spending a significant amount of time on regression analysis. The second unit will carry forward the notion of uncertainty presented in the first unit, and focus on modeling techniques for using data to make better decisions under uncertainty. We will emphasize interpreting model results and will spend two weeks working on simulation approaches. The final unit will introduce optimization models, with the objective of providing an understanding of the capabilities of optimization and recognizing opportunities to deploy such models.

To allow students to focus on the key concepts of the course, the course is built around the Radiant Tool. Radiant is a data processing and visualization tool that does not require coding, while operating within the R environment (which is free to use).

## OBJECTIVES

At the close of this course, you will be able to:

- Read and interpret business analytics reports
- Produce data visualizations using Radiant software
- Understand how to use data to improve operations and decision making
- Build a wide array of models to help make decisions under uncertainty
- Synthesize and communicate the important findings from modeling and data analysis

## MATERIALS

Required:

Data, Models, and Decisions: The Fundamentals of Management Science, by Dimitris Bertsimas and Robert M. Freund, 2004, Dynamic Ideas, Belmont, MA, ISBN 0-9759146-0-X.

### SCHEDULE

| Date                | Due                    | Class Topic                                 |
|---------------------|------------------------|---------------------------------------------|
| Week 1 September 29 |                        | Probability                                 |
| Week 2 October 6    | Assignment 1           | Hypothesis Testing and Confidence Intervals |
| Week 3 October 13   | Assignment 2; Case 1   | Regression Analysis Part I                  |
| Week 4 October 20   | Assignment 3           | Regression Analysis Part II                 |
| Week 5 October 27   | Assignment 4; Case 2   | Decision Analysis                           |
| Week 6 November 3   | Midterm Project Report | Midterm Project Presentations               |
| Week 7 November 10  | Assignment 5; Case 3   | Simulation Part I                           |
| Week 8 November 17  | Assignment 6           | Simulation Part II                          |
| Week 9 November 24  | Assignment 7; Case 4   | Linear Optimization Models                  |
| Week 10 December 1  | Assignment 8; Case 5   | Additional Topics & Review                  |

### ASSIGNMENTS

There will be eight individual assignments, five group cases, and a midterm project.

The eight individual assignments will be submitted as quizzes in Canvas. You will have the opportunity to take each of these quizzes twice. This will allow you to see which questions you got wrong on the first attempt and retry these questions for learning purposes. You must include your handwritten and/or Radiant work as part of the submission. Teaching assistants will check to ensure students have submitted this work. If you collaborated with classmates, please indicate who you worked with at the top of your work. All students must submit their own work.

The five group cases will require more in-depth analysis and provide an opportunity to practice communicating your analyses and findings. We will cover expectations and examples for these assignments in the first week of the course.

The midterm project will be an opportunity for you and your group to conduct an in-depth analysis of a dataset, write a report of your findings, and present a summary to the class. The presentations will take place during week 6. You will be expected to incorporate the topics covered in weeks 1-4 in the report. The instructor will provide several datasets that students can use, but encourages students to find their own datasets that interest them, if this option is available to them. More information will be provided in the first two weeks of the class.

### READING ASSIGNMENTS

| Date                |                                 |
|---------------------|---------------------------------|
| Week 1 September 29 | Sections 2.1, 2.4-2.7, 3.1, 3.4 |
| Week 2 October 6    | Sections 3.7, 4.1-4.3, 4.7, 4.8 |
| Week 3 October 13   | Sections 3.7, 4                 |
| Week 4 October 20   | Sections 2.9, 6.1-6.4           |
| Week 5 October 27   | Section 1                       |
| Week 6 November 3   | None                            |
| Week 7 November 10  | Sections 5.1-5.3, 5.7-5.9       |
| Week 8 November 17  | None                            |
| Week 9 November 24  | Sections 7.1-7.6                |
| Week 10 December 1  | None                            |

These reading assignments should be completed prior to each class listed above. There will also be Radiant Tutorials within the modules Canvas page to be completed before class in certain weeks.

### EXAM

The final exam will take place on Monday, December 8 from 11:30am-2:30pm in OTRSN 4E106. The exam will be paper-based and test key concepts, especially your ability to interpret the results of data analysis. A sheet of formulas and definitions will be provided in advance and can be used during the exam.

### GRADING

| Assignments                     | Percentages |
|---------------------------------|-------------|
| Final Exam                      | 40%         |
| Midterm Project & Presentation  | 25%         |
| Individual Homework Assignments | 20%         |
| Group Case Assignments          | 15%         |
| Total                           | 100%        |

### CHATGPT AND OTHER AI TOOLS

ChatGPT and other AI tools are now widely available. In your careers you will likely have to figure out when these tools are helpful and when they are not. Having said this, as this is a class meant to help you develop a strong foundation for your MBA education, you are discouraged from using these tools for anything other than simple searches within this class. If you do use ChatGPT or other AI tools, you must cite all instances of use beyond simple searches. Any cases where it appears AI tools were used and not cited will be referred for academic integrity violations. When in doubt, cite.

### ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at:

<http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

### STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or [osd@ucsd.edu](mailto:osd@ucsd.edu).